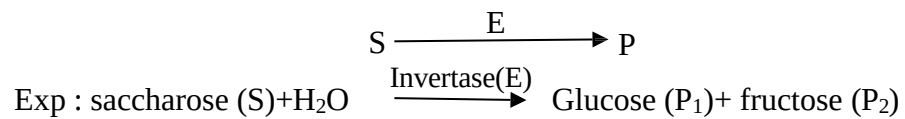
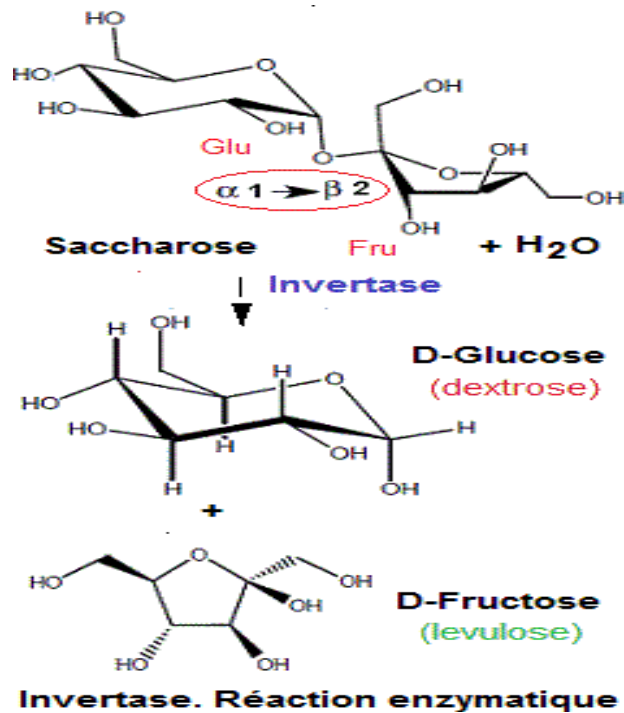


Introduction to Enzymology

Introduction

All metabolic reactions are catalyzed by enzymes, also known as biocatalysts or catalytic proteins. The compound transformed by an enzyme is called the substrate, and the resulting compound is called the product. Enzymes (E) are proteinaceous compounds that catalyze biological reactions in which a substrate (S) is converted into a product (P).





1. . Properties of Enzymes

➤ Efficiency

Enzymes are extremely efficient; for example, OMP decarboxylase increases the rate of a reaction by a factor of 10^7 .

➤ Non-consumption

At the end of the catalytic process, enzymes are recycled (regenerated) and therefore remain intact at the end of the reaction.

➤ Acts at low concentration

Due to their efficiency and non-consumption at the end of the reaction, enzymes act at very low concentrations (or amounts) compared to the substrate concentration.

➤ **Specificity**

Enzymes exhibit high specificity toward their substrates.

➤ **Acts at low temperatures**

Because of their proteinaceous nature, enzymes function at relatively low temperatures.

enzymes régulatrices.

2. Structure of Enzymes

Enzymes are proteins, meaning they are polymers of amino acids forming long chains folded in space. They have a compact structure folded onto itself, which is necessary for catalytic activity.

Enzymes exhibit several levels of organization:

- **Primary structure:** The sequence of amino acids linked together by peptide bonds. This structure corresponds to a linear chain without any particular spatial organization.
- **Secondary structure:** The amino acid sequence folds to form motifs, such as α -helices and β -sheets.
- **Tertiary structure:** The fully folded structure corresponds to the native state; the protein adopts a compact form (known as a globular protein) stabilized by numerous bonds (hydrogen, ionic, hydrophobic interactions, and disulfide bridges). This shape allows enzymatic activity (see Table 1).
- **Quaternary structure:** For some enzymes, an additional level of organization exists, where multiple protein chains associate. This structure is typically found in regulatory enzymes.

Table 1 : structure of enzymes

Structure	Primaire	Secondaire	Tertiaire	Quaternaire
	...-Ala-Glu-Val-Thr-Asp-Pro-Gly- chaîne polypeptidique	 hélice α  feuillet β		
Définition	Ordre d'enchaînement des acides aminés (=séquence des aa)	en raison des liaisons peptidiques : - Hélice α : enroulement de la chaîne d'aa à la manière d'un ressort - Feuillet β : disposition parallèle des chaînes d'aa, en zigzag, les replis s'effectuant au niveau des carbones α	Association d'hélices α et de feuillets β en une super-structure tridimensionnelle	Association de plusieurs sous-unités protéiques de structure tertiaire.

1.1. Enzyme Active Site

The function of enzymes is linked to the presence of a specific region in their structure called the **active site**. Schematically, it has the shape of a cavity or groove in which substrates bind through several weak chemical interactions. Once bound, the substrates undergo a reaction and are converted into products.

The active site is divided into two parts:

- **Binding/recognition site:** Recognizes the shape complementarity with a substrate specific to the enzyme.
- **Catalytic site:** Facilitates the reaction that transforms the substrate into the product (see Figure 1).

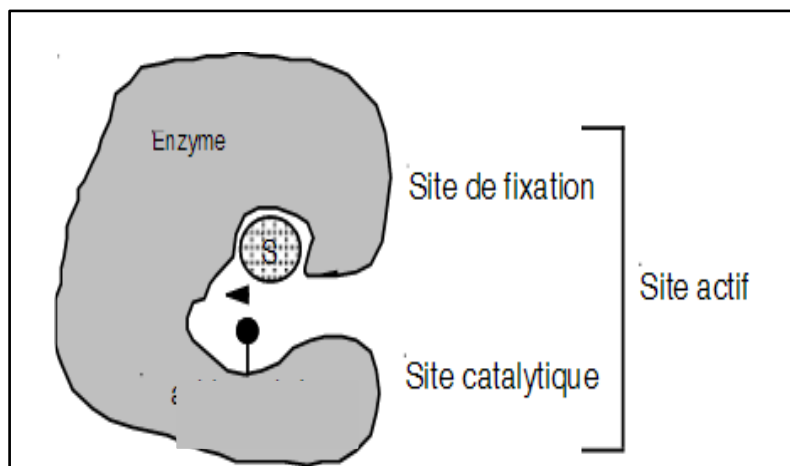


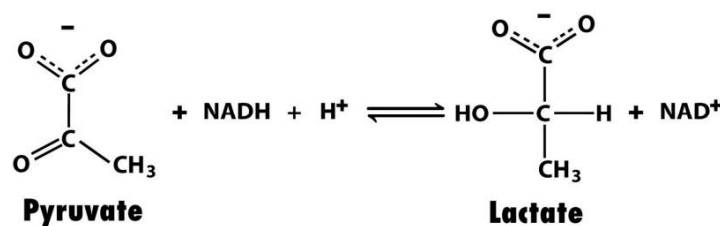
Figure 1 : Representation of the Active Site

Note:

Some enzymes are composed of a protein part called the **apoenzyme** (thermolabile) and a non-protein part called a **cofactor** or **coenzyme** (thermostable). The association of these two parts forms the **holoenzyme**.

1.1.1. Cofactors

A **cofactor** is a non-protein chemical compound that binds tightly to an enzyme, contributing to its function. It attaches to the inactive form of the enzyme, called the **apoenzyme**, thereby activating the enzyme. Enzymes often require cofactors, which are essential for the reaction to proceed. They are frequently derived from vitamins.



Enzyme : lactate déhydrogénase



A coenzyme is a non-protein organic compound that binds to an enzyme to help catalyze a reaction.

Coenzymes are a type of cofactor; however, there are distinctions between them. The key difference is that coenzymes are organic molecules, whereas cofactors can be either organic or inorganic molecules.

2. The Enzyme-Substrate Complex

The progress of a chemical reaction depends not only on the reactivity of the components involved (the enzyme and the substrate) but also on their encounter and affinity. The same principles apply to a biochemical reaction, with the significant advantage that substrates (S) are held at the enzyme's (E) active site to form the enzyme-substrate complex (ES). This complex then leads to the enzyme-product complex (EP) and ultimately to the release of the product (p).

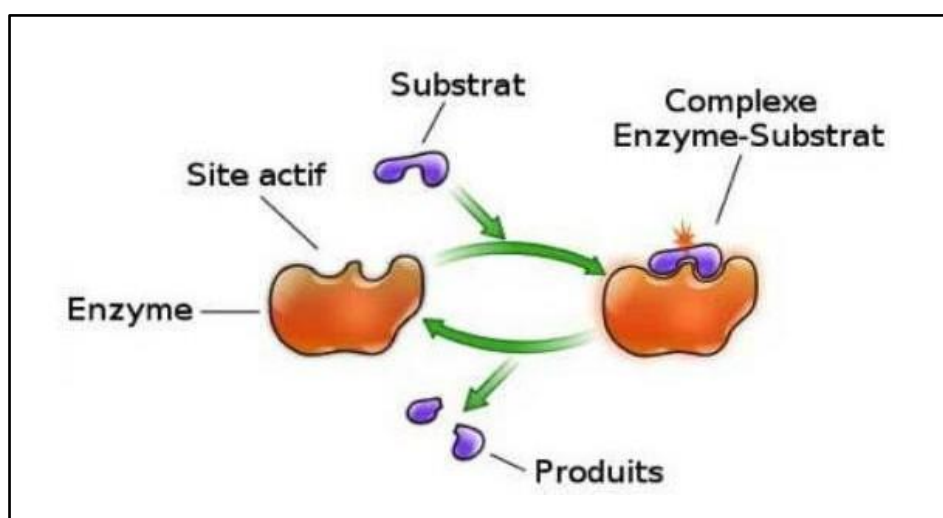


Figure 2 : Schematic representation of an enzymatic reaction.

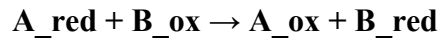
2. Nomenclature and Classification of Enzymes

A number of enzymes are commonly referred to by traditional names established by usage (pepsin, trypsin, chymotrypsin, papain, etc.), which provide little information about the substrate or the reaction catalyzed. In some cases, to name enzymes that catalyze hydrolysis reactions, the name of the substrate is used followed by the suffix “-ase” (e.g., peptidase, phosphatase). A slightly more precise naming system involves using the name of the substrate followed by the reaction catalyzed with the suffix “-ase”; for example, **malate dehydrogenase**.

In 1961, the **Enzyme Commission (E.C.)** of the International Union of Biochemistry established a systematic classification and nomenclature, much more rigorous, based on the biochemical reaction catalyzed by the enzymes. The six classes are:

- **Oxidoreductases**, which catalyze oxidation-reduction reactions (e.g., peroxidase).

This first class includes enzymes that catalyze oxidation or reduction reactions, whose most general form can be written as:



These reactions therefore involve the transfer of electrons, hydrogen atoms, or the incorporation of oxygen. Any oxidation corresponds to the removal of one or more electrons from a substrate. These electrons are transferred to another molecule, the acceptor, which is consequently reduced. These enzymes necessarily have two substrates and require the presence of a coenzyme (NAD, NADP, FAD).

- **E.C.2: Transferases**, which transfer a functional group from one molecule to another (for example, methyltransferases, which transfer a methyl group);
- **E.C.3: Hydrolases** are degradative enzymes that cleave a molecule and attach the H and OH radicals from water to the freed valences. These are enzymes without coenzymes. They act on ether, acetal, ester, and phosphate groups, O–O bonds in peroxides, C–N bonds in amides, etc.
- **E.C.4: Lyases or synthases**, which break bonds but simultaneously form new ones (like adenylate cyclase, which produces cyclic AMP from ATP).
- **E.C.5: Isomerases**, which rearrange the functional groups of a molecule to form isomers (like topoisomerases that coil DNA).
- **E.C.6: Ligases or synthetases**, which enable the joining of two molecules (like DNA ligases).

3. The official nomenclature (code number)

In the so-called "official" nomenclature, the classification of enzymes is based on specificity criteria. An enzyme is assigned a code number, preceded by the letters E.C. (Enzyme Commission) and consisting of 4 digits separated by periods. The enzyme is generally written in the form: E.C. X1.X2.X3.X4

- **X1:** Ranges from 1 to 6 and indicates the six types of reactions catalyzed by enzymes (class).
- **X2:** Specifies the type of donor group or substrate on which the enzyme acts (subclass).
- **X3:** Specifies the type of acceptor group for protons/electrons on which the enzyme acts (sub-subclass).
- **X4:** Indicates the serial number of the substrate on which the enzyme acts (registration number of the enzyme within the relevant sub-subclass).

Example: Alcohol dehydrogenase: E.C.1.1.1.1

- **1:** Oxidoreductase
- **1.1:** Acts on CH–OH groups
- **1.1.1:** With NAD or NADP as acceptor
- **1.1.1.1:** First enzyme discovered and recognized



4. Measurement of Enzymatic Activity in Human Pathology

To investigate a disease, enzymatic tests are frequently performed in medical laboratories. Indeed, during cellular distress, enzymes that are normally intracellular can be released into the plasma at high concentrations.

- Heart cells secrete proteins and cardiac enzymes when the heart is under stress. The main protein used in medicine for diagnosing cardiac distress is troponin. This enzyme is released early, for example, during a myocardial infarction, and a high level requires urgent treatment of the underlying cause. Certain enzymes are measured during a myocardial infarction, such as glutamate-oxaloacetate transaminase, among others.

- An increase in liver enzymes in the blood indicates an abnormality in the liver or bile ducts. Elevated transaminases (ALAT and ASAT) reflect cytolysis (destruction of liver cells, accompanied by a parallel rise in the enzymes measured in the blood), for example in cases of viral hepatitis, alcoholism, drug toxicity, or exposure to other substances.
- An increased level of lipase (a pancreatic enzyme) may indicate damage to the pancreas or intestines, and it is usually necessary to request the measurement of additional parameters.
- Concurrent increases in digestive enzymes (amylase, lipase, etc.) can indicate a disease of the **pancreas** or salivary glands.

Références bibliographiques :

1-N. Latruffe, 2014. Biochimie : tout le cours en fiche, édition Dunod, 107p.

2- J.H. Weil, 2005. Biochimie générale, édition Dunod, 66p.